

① Problem: Compute the eigen values and eigenvectors of $X = AA^T$ where $A \in \mathbb{R}^{m \times d}$, $m \gg d$, without directly performing eigendecomposition on the large $m \times m$ matrix.

Solution:

Suppose $u \in \mathbb{R}^m$ is an eigenvector of X with eigenvalue

$$Xu = \lambda u$$

$$\Rightarrow AA^T u = \lambda u$$

Multiply with A^T both sides

$$A^T A A^T u = A^T \lambda u$$

$$\Rightarrow (A^T A)(A^T u) = \lambda(A^T u)$$

Thus $A^T u$ is either zero vector or an eigenvector of the smaller matrix ($d \times d$) with same eigen value.

Instead of solving the eigen problem for $X (m \times m)$, we solve it for $A^T A \in \mathbb{R}^{d \times d}$. Since $d \ll m$, this computation is much cheaper.

Let (v, λ) be an eigen pair for $A^T A$

$$A^T A v = \lambda v$$

Lift eigenvectors back to the big Matrix

multiply both sides by A

$$AA^T(Av) = \lambda(Av)$$

Thus $u = Av$, which is an eigenvector for matrix X with eigen value λ .

Finally normalize u to unit length

$$u = \frac{Av}{\|Av\|}$$

Algorithm:

- 1) Form the $d \times d$ matrix $M = A^T A$
- 2) Compute its eigen decomposition $Mv_i = \lambda_i v_i$
- 3) For each eigenvector v_i , construct the eigenvector of X as $u_i = \frac{Av_i}{\|Av_i\|}$
- 4) The corresponding eigenvalues of X are ^{the} same λ_i

Conclusion:

By solving the eigen problem of the smaller $d \times d$ matrix $A^T A$, we can efficiently obtain the eigenvalues and eigenvectors of the large $m \times m$ $X = AA^T$

②

(a) True

Linear regression is a supervised learning method. It uses labelled data (x_i, y_i) to learn a function $f(x) \approx y$.

(b) True

PCA is an unsupervised learning method. It analyzes the input features X , with no labels, in order to find structure in data. ~~Spec~~

Specifically, it identifies the directions (principal components) along which the variance is maximized.

False

The statement is reversed.

SVD is a general linear algebra operation on any matrix, decomposing it into orthogonal factors.

$$A = U \Sigma V^T$$

PCA is specifically applied to a dataset: it computes the covariance matrix of the features and performs eigen decomposition (or equivalently, SVD of the centered data matrix).

Thus PCA is primarily a technique applied to a dataset, while SVD is primarily a technique applied to matrix.

(d) Multicollinearity arises when input features are linearly dependent (eg height in inches and centimeters). This makes $X^T X$ nearly singular, producing unstable regression coefficients.

A standard preprocessing fix is to use PCA before regression. PCA replaces correlated features with orthogonal principal components, ensuring the inputs are linearly independent. This removes redundancy and

Stabilizes the regression problem.

So, we need to use PCA to remove multicollinearity.